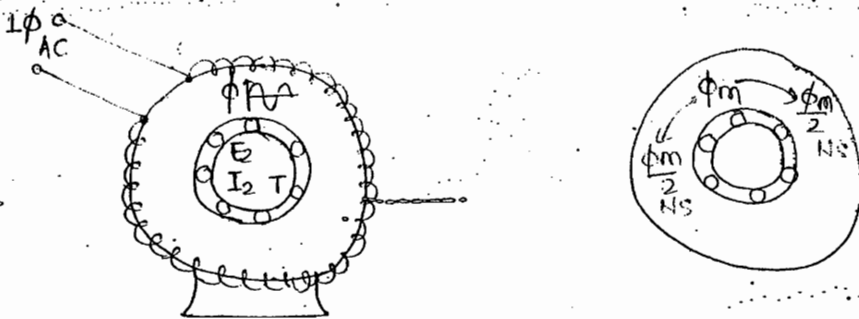


* When rotor fed the absolute flux in space rotates at an speed N_s wrt. rotor & runs at slip speed $(N_s - N)$ wrt. stator (or) stationary point at space.

(d) $\frac{26}{28}$ (b) $\frac{77}{9}$

Single phase Indⁿ motors

- * If the fuse blows off in 3ϕ motors when running at light weight condⁿ it won't stop but continue to run as 1ϕ motor with reduced speed & increased current.
- * 1ϕ motor are popular because of there small ratings 1w to 1kw & domestic supply is 1ϕ .
- * There construction is as identical to 3ϕ motors.
- * stator contain 1ϕ wdg for 2, 4, 6 no. of poles generally.
- * Depending on the starting method adopted stator wdg differs.
- * Rotors are essentially sq. cage type with skewing.
- * The basic limitation of 1ϕ motors compare to 3ϕ is due to there non-self starting nature which can be analysed acc. to double field revolving theory. The complete analysis, eq. ckt representation is based on this theory.



When a 1ϕ supply is applied across stator containing one wdg a pulsating (or) alternating magnetic field is produced.

It links with rotor & results in induced emf as the rotor is essentially closed it develops current & torque but rotor doesn't start.

If an external force applied the rotor continues to rotate in the dirⁿ of force & reaches its rated speed.

This behaviour is analysed acc. to double field revolving theory.

Acc. to this theory a pulsating magnetic field contains 2 magnetic fields running at synchronous speed of equal magnitude ($\phi_m/2$) & in opposite dirⁿ.

Forward: ϕ_f $\phi_m/2$ N_s T_f N $s_f \Rightarrow s_f = \frac{N_s - N}{N_s} = s$

Backward: ϕ_b $\phi_m/2$ N_s T_b $-N$ $s_b \Rightarrow s_b = \frac{N_s + N}{N_s} = \frac{N_s + N - N_s + N_s}{N_s} = \frac{2N_s - (N_s - N)}{N_s}$

$= 2 - s$

* As the rotor is subjected to both fields the entire operating region of 1 ϕ motor will be considered b/n 2 extreme boundary condⁿ 0 & 2.

At start $s = 1$

$s_f = 1, s_b = 1$

motoring region for 1 ϕ IM

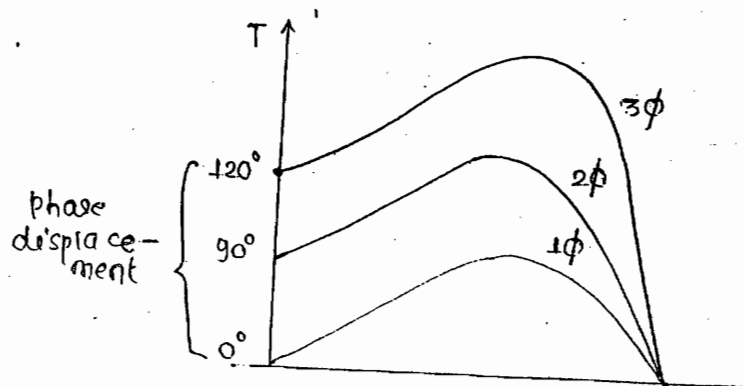
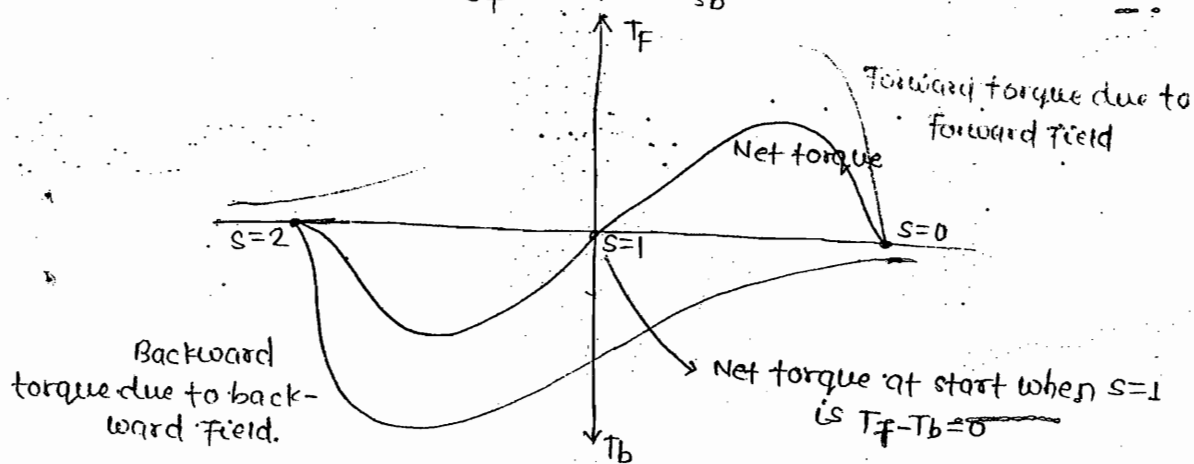
Running $s = 0$

$s_f = 0, s_b = 2$

$s = (0, 1, 2)$

$T_g = \frac{60}{2\pi N_s} \times \frac{RCL}{s}$

$T_f \propto \frac{I^2 R_2}{s_f}$; $T_b \propto \frac{I^2 R_2}{s_b}$



* RMF with 2- ϕ supply $\rightarrow$

$$\phi_A = \phi_m \sin \omega t$$

$$\phi_B = \phi_m \sin \omega t$$

90°

at $\omega t = 0$; $\phi_A = \phi_m$,

$\phi_B = 0$

ϕ_m

at $\omega t = 45^\circ$; $\phi_A = \phi_m / \sqrt{2}$

$\phi_B = \phi_m / \sqrt{2}$

$\phi_m / \sqrt{2}$

45°

$\phi_{net} = \phi_m$

at $\omega t = 90^\circ \rightarrow \phi_A = 0$

$\phi_B = \phi_m$

ϕ_m

ϕ_m

N_s

Rotate in cw dirn

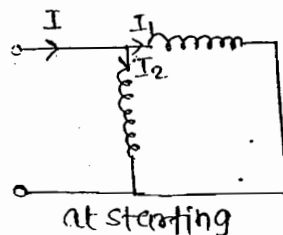
If we want to change the dirn of field the reverse the phase seq.

$\phi_A = \phi_m \sin \omega t$

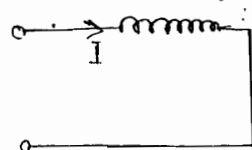
N_s

$\phi_B = \phi_m \cos \omega t$

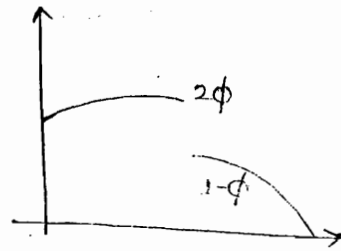
* Like a 3 ϕ supply a 2 ϕ supply also produce RMF which rotates at syn. speed but with reduced strength ϕ_m only.



split 1 ϕ into 2 parts & connect parallel.



at running

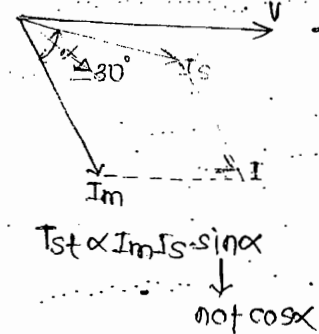
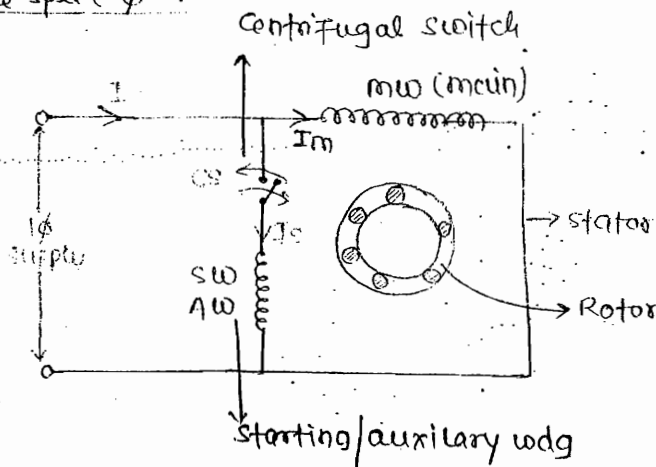


* Depending on starting method adopted 1- ϕ motors are classified into 2 types:-

- (1) Split-phase
 - Resistance split phase/split phase
 - capacitor start
- (2) Split-pole
 - Capacitor start & capacitor run/capacitor run motor.

↓
shaded pole.

* Resistance split- ϕ →



* This is most popular for motors rating around 250w only.

* Stator contains split ϕ wdg (2 wdg):-

(i) main wdg (running wdg) which is having
 (a) large no. of turns more inductive naturally

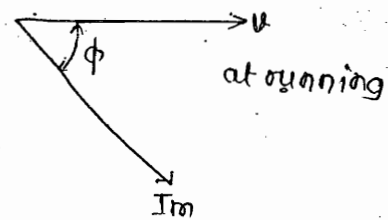
(ii) Starting (or) auxiliary wdg which is having less

no. of turns intentionally made highly resistive with thin cond^r.

* This will introduce an angle of displacement b/n both current in wdg

I_m & I_s around $\alpha = 30^\circ$.

* It is sufficient to produce starting torque for small rating motors around 250w but not suitable for high inertia loads above 250w ratings.



* This are not suitable for frequent starting & stopping but for continuous operation in its segments. α is low current drawn high.
(Because of temp. rise in the resistor).

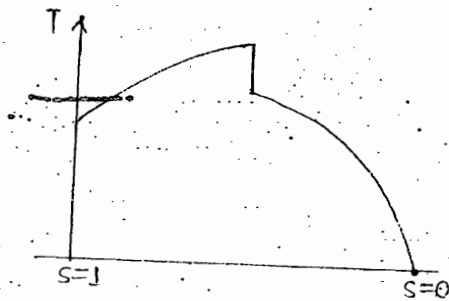
Appln →

250W
Domestic / Sump motors

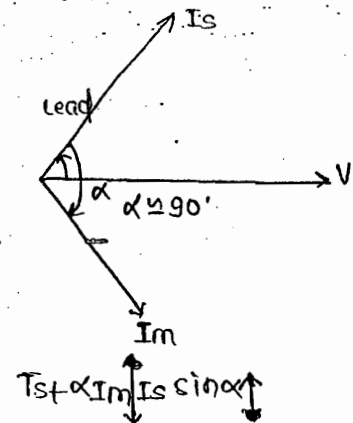
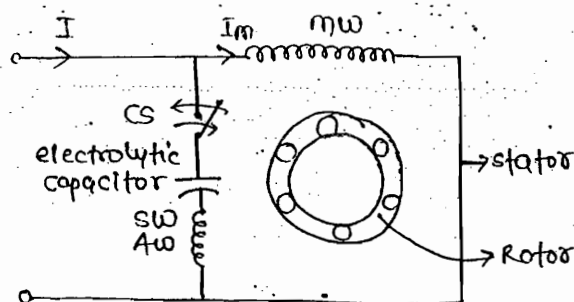
Centrifugal motors.

Fans / blowers.

* Once the motor starts & reach 80% of rated speed the centrifugal switch gets opened to isolate starting wdg from supply. & motor continues to run with main wdg only.



(2) Capacitor start →



* In order to create a large ϕ diff b/n I_m & I_s ($\alpha \approx 90^\circ$) a capacitor is included in series with starting wdg through a switch.

* As α is high the starting current drawn is comparatively low.

* It has high starting torque & used for high inertia load (hard to start) of 250-750W.

* As the motor reaches near rated value of speed the centrifugal switch will get

Opened & ~~this~~^{dis}connects both capacitor & starting wdg.

* Under running condⁿ it is quite similar resistance split phase starting.

This is preferred for frequent startings & stopping load specially.

Applications →

250-750W

Fans/blowers

sump pumps/centrifugal pump

refrigerator units, AC, washing m/c, grinders etc.

Electrolytic capacitor → cost effective

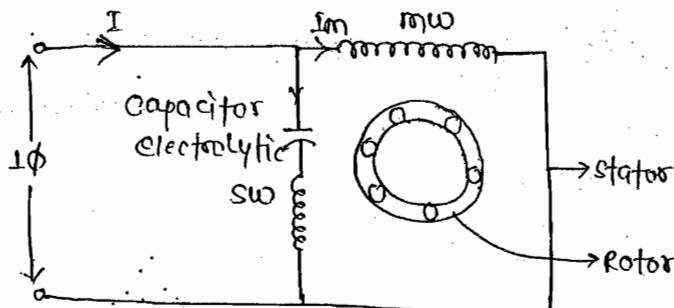
(3) Capacitor start & capacitor Run →

* There is no centrifugal switch & the capa. is continuously connected to supply along with starting wdg.

* The design cost is high comparatively because the starting wdg should be efficient.

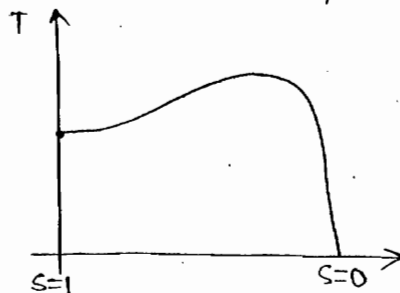
* It starts & runs as perfect 2 ϕ motor but capacitor start motor starts as 2 ϕ motor but runs as $\pm\phi$ motor with low PF.

* Capacitor run motors are popular because of their silent operation with smooth T-s c/s & good starting as well as running torques.

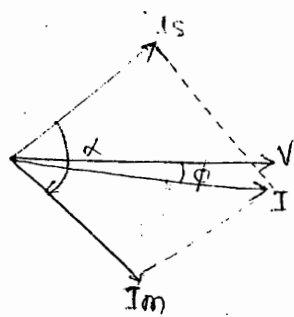


Appliⁿ →

All ceiling fans, AC & motors in (fans/blowers/pumps) of ratings 750-1000W



* As the capacitor is continuously connected 2 small range capacitor are used in parallel with switching option externally, depending starting & running torque they can be switched.

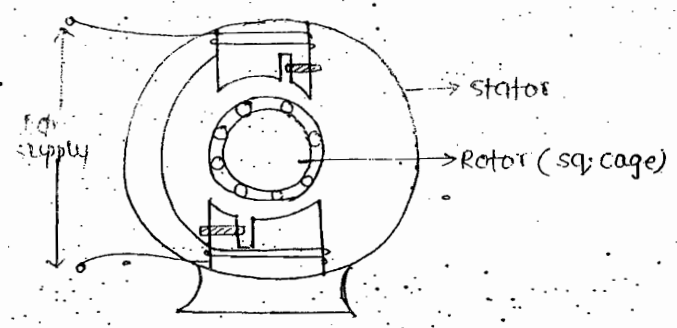


angle low
PF High

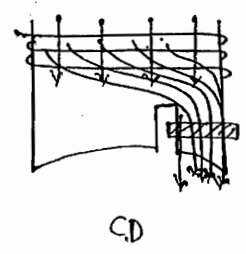
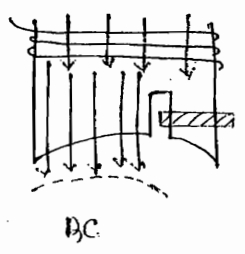
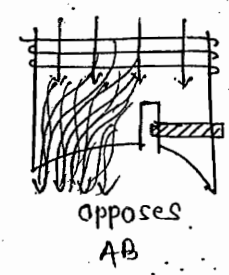
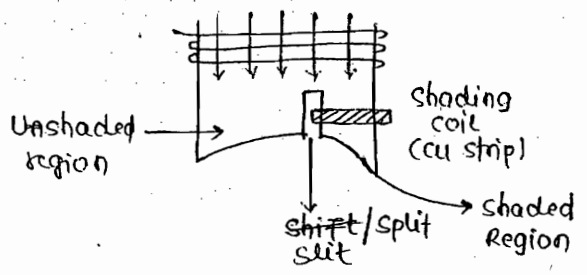
$$P = VI \cos \phi$$

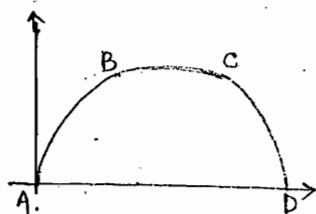
$$\uparrow \cos \phi = \frac{P}{VI}$$

* Shaded pole $\rightarrow$



[-50w Ranges
Toys]





Application →

Toys.

Small Fans/ Desk Fans

Cooling Fans in laptop

Xerox m/c

Advertisement displays.

Compact appliances like Hair dryers etc.

* It is popular for its compactness & widely used in small ranges less than 50W.

* Stator is totally different from split phase type.

* It contains projected poles with a split & cu strip. The solidly closed in one region of pole called as shaded region.

* At instant AB the rate of change of flux linkage is +ve, EMF induces in the shading coil produce current & its own flux to oppose the main flux coming out of the pole.

Therefore all the flux lines gets crowded in unshaded region at instant

* BC rate of change of flux linkage is very low & EMF induced current & flux of shading coil negligible.

Therefore the flux distributes uniformly through out the pole.

* At instant CD Rate of change of flux linkage are in the -ve manner.

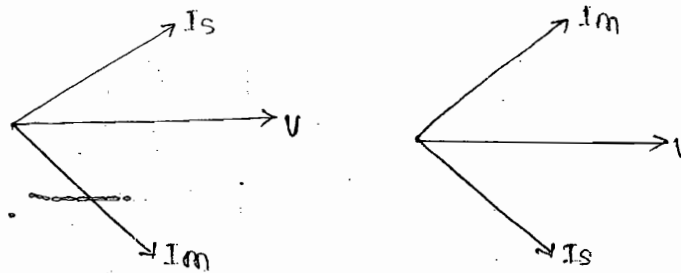
Therefore the opposition will be in +ve manner which makes the flux lines crowded in shaded region.

* This will create a moving or RMF in air gap under all the poles.

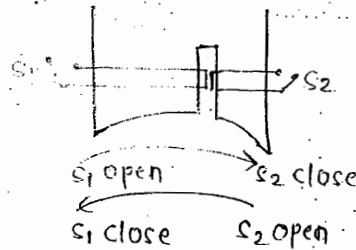
However starting torques are very low & non-uniform magnetic field results in the non-uniform speeds & torque.

* In order to reverse the dirn of rotation of split phase motor:-

- (1) Either reversing mw terminals.
- (2) Either reversing sw terminals.
- (3) Interchanging capacitor from starting wdg to main wdg.



* By reversing both mw & sw (or) reversing supply terminals doesn't change the dirn of rotation of motor.



* The flux always moves from unshaded to shaded region only.

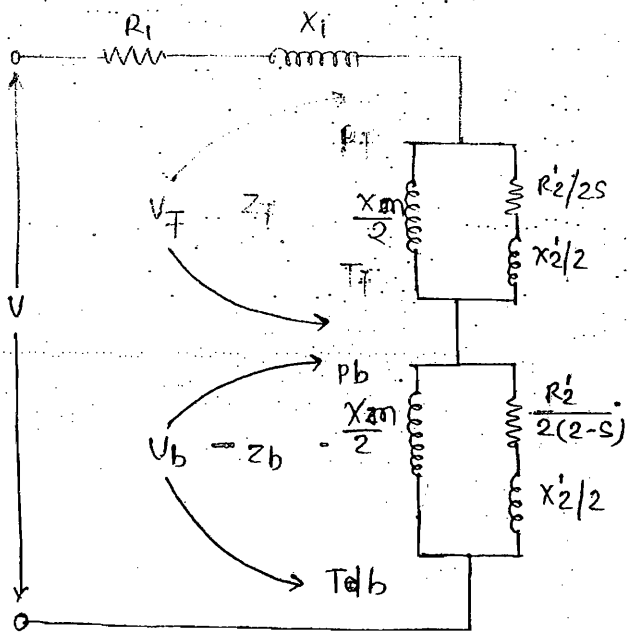
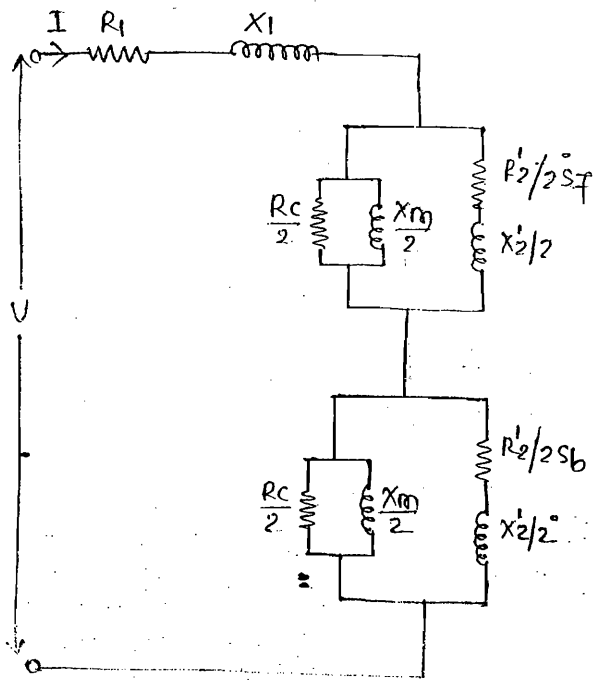
Therefore it requires 2 shading coils which can be externally switch.

* Equivalent circuit representation →

* The motor is represented on eq. ckt for further analysis.

* In 1 ϕ motors the representation acc to double field revolving theory.

* Acc. to this as one rotor is reacting to both forward & backward fields it will be one stator 2 rotor representation as shown below.



$$* Z_1 = R_1 + jX_1$$

$$* Z_f = j \frac{X_m}{2} \parallel \left(\frac{R_2}{2s} + j \frac{X_2}{2} \right)$$

$$= R_f + jX_f$$

$$* Z_b = j \frac{X_m}{2} \parallel \left(\frac{R_2}{2(2-s)} + j \frac{X_2}{2} \right)$$

$$Z_{eq} = Z_l + Z_f + Z_b$$

$$I = \frac{V}{Z_{eq}}$$

$$I \angle \phi \quad \therefore PF = \cos \phi \text{ (lagg)}$$

Power i/p to forward rotor $\rightarrow I^2 R_f = P_f$

Power i/p to backward rotor $\rightarrow I^2 R_b = P_b$

Net P_{i/p} to rotor (OR) Air gap power = $P_f - P_b$

Power o/p of rotor = $(1-s)(P_f - P_b)$

* For accuracy in the analysis from eq. circuit while calculating net power call iron, friction & windage (const. or rotational losses) should be included while eliminating shunt branch resistance R_c from actual ckt

Net/shaft/motor o/p = $(1-s)[P_f - P_b] - \text{Rotational loss}$

$$\eta = \frac{\text{Shaft o/p}}{E \cdot \text{Input}} = \frac{(1-s)[P_f - P_b] - [\text{Rotational loss}]}{V_1 \cos \phi}$$

$$\text{Gross torque } T_g = \frac{60}{2\pi N_s} \text{ Rot. P}_{i/p}$$

$$T_g = \frac{60}{2\pi N_s} (P_f - P_b)$$

$$\text{Shaft torque } T_{sh} = \frac{60}{2\pi N} \text{ shaft o/p}$$

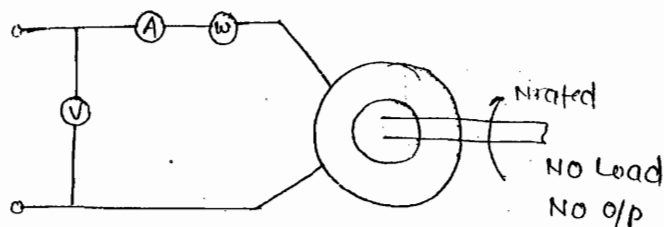
$$T_{sh} = \frac{60}{2\pi N} P_{sh}$$

$$\text{Shaft o/p} = P_{sh} = (1-s)[P_f - P_b] - [\text{Rotational loss}]$$

* Determination of eq. ckt parameter →

It requires determination of 3 tests to be conducted in order to find out the parameters of eq. ckt as similar to 3 ϕ motors.

(1) No Load test →



* Apply rated vol. across the given 1 ϕ Im & allow it to run freely on NL.

$$I_p = \text{o/p} + \text{loss (total)}$$

$$I_p = \text{loss}$$

- stator cu loss
- stator core loss
- Rotor cu loss $\times$ (current is low)
- Rotor core loss $\times$ (sf is small)
- mech. loss

* It requires 1 ϕ wattmeter to measure NL power i/p.

* The wattmeter reading is considered to be containing stator core loss + mech. loss + small stator cu loss.

(2) DC test →

* Apply small vol. (dc) across the stator by connecting a v-m & ammeter.

* Ratio of V/I gives dc resistance.

* It is adjusted to ac value by multiplying with suitable factor around 1.5.

$$R_{1Ac} = 1.5 R_{Dc}$$

$$W_0 - I_0^2 R_1 = \text{Rotational loss (iron, friction \& windage)}$$

* From stator resistance value rotational loss is calculated.

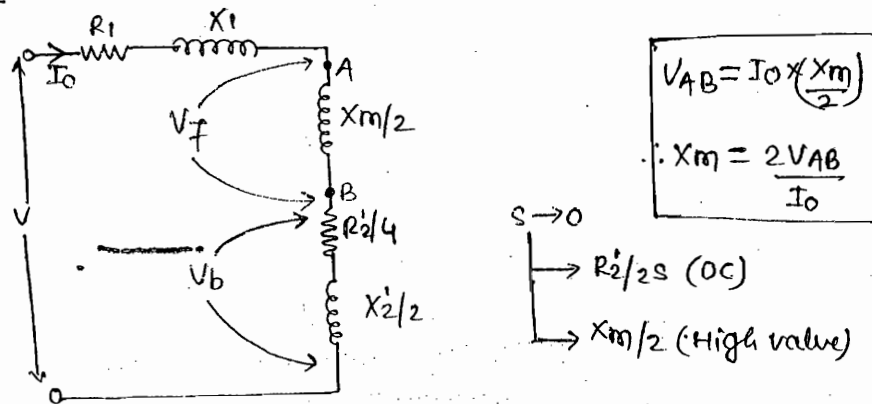
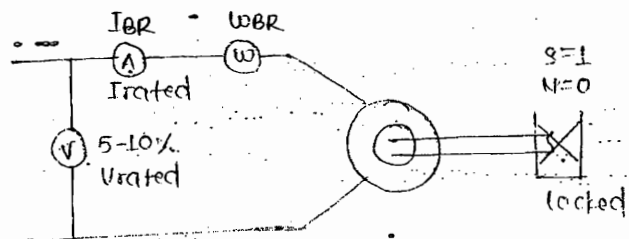
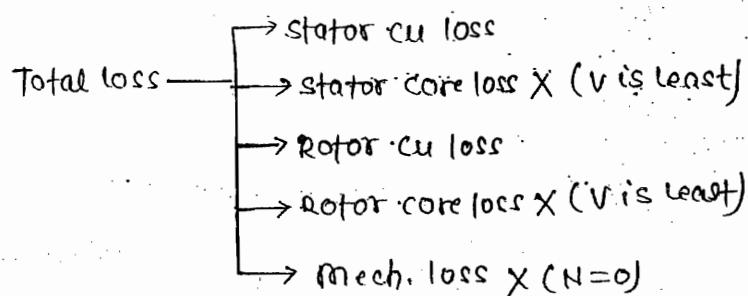


Fig:- Eq. ckt at No Load

(3) Blocked Rotor test $\rightarrow$

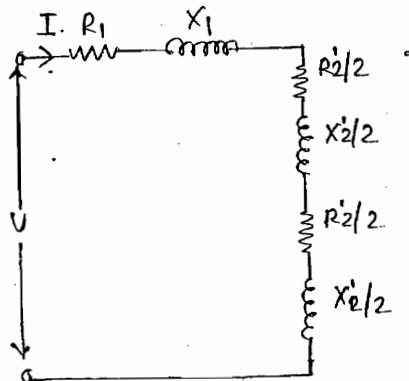


* Rotor is initially blocked & small vol. is applied across stator through a suitable auto Xmer while insuring rated current drawn by motor.



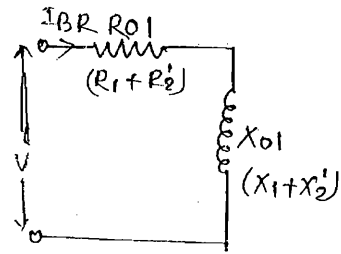
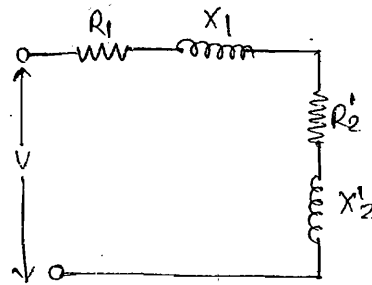
wattmeter reading $\rightarrow$ only cu loss

* In this test wattmeter reading represent both stator & rotor cu losses.



$s=1$

Fig:- Eq. ckt on blocked rotor condⁿ



$$W_{BR} = (\text{stator} + \text{rotor}) \text{ cu loss}$$

$$= (I^2 R_1 + I^2 R_2') = I^2 R_{01}$$

$$= I_{BR}^2 \cdot R_{BR}$$

$$W_{BR} = I_{BR}^2 \cdot R_{BR}$$

$$R_{BR} = \frac{W_{BR}}{I_{BR}^2}$$

$$Z_{BR} = \frac{V_{BR}}{I_{BR}}$$

$$X_{BR} = \sqrt{Z_{BR}^2 - R_{BR}^2}$$

$$X_1 = X_2' = \frac{X_{BR}}{2}$$

$$X_{BR} = X_{01} = X_1 + X_2'$$

$$Z_{BR} = Z_{01}$$

$$\eta = \frac{\text{O/p}}{\text{O/p} + \text{Const} + \frac{\text{Cu}}{\text{loss}}} = \frac{\text{I/p} - \left[\frac{\text{const}}{\text{loss}} + \frac{\text{Cu}}{\text{loss}} \right]}{\text{I/p}}$$

* Comparison b/w 3 ϕ & 1 ϕ Im $\rightarrow$

* 3 ϕ power capacity is more than that of 1 ϕ .

* For a same rating 1 ϕ motors are comparatively having more size, cost, low torque & run with a noise due to backward field.

Cost also comparatively more in 1 ϕ because of Auxiliary wdg, switch & capacitor.

* Comparison b/n

* 3 ϕ motors are more efficient than 1 ϕ .

$$T \propto \frac{SV^2}{R_2 f}$$

(20/27)

(b)

$$\downarrow T \propto \frac{SV^2}{R_2 f}$$

(c)

$$\uparrow T \propto \frac{SV^2}{R_2 f}$$

(d)

$$\uparrow T \propto \frac{SV^2}{R_2 f}$$

(28/28)

$$T \propto SV^2, \quad \frac{T_2}{T_1} = \left(\frac{S_2}{S_1}\right) \left(\frac{V_2}{V_1}\right)^2$$

(29/28)

$$S_m = \frac{1200 - 1000}{1200} = \frac{1}{6}$$

(30/29)

slip ring $\rightarrow$ No. of stator pole =

No. of rotor pole

$$S_m = \frac{1}{6} = \frac{R_2}{X_2}$$

$$X_2 = 7.2$$

$$R_2 = X_2 \text{ (max}^m \text{ torque)}$$

$$X_2 = R_2 + R'_2 \text{ (series add)}$$

$$R'_2 = 7.2 - 1.2 = 6 \Omega$$

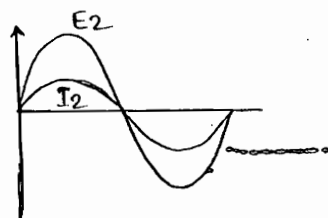
(31/30)

* Commutator converts AC $\rightarrow$ DC, therefore in the extra load freq. is 0.

* If there are slip rings the freq. across them is slip freq.

* freq. wrt brushes on the commutator will be supply freq. (or) stator freq. only.

(37/30)



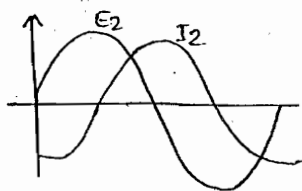
Pure Resistive

$$\phi = 0$$

$$\cos \phi = 1$$

$$T \propto E_2 I_2 \cos \phi_2$$

$$T \propto F_1 F_2 \sin \delta$$

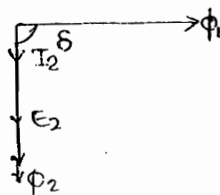
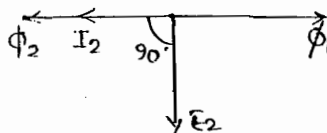


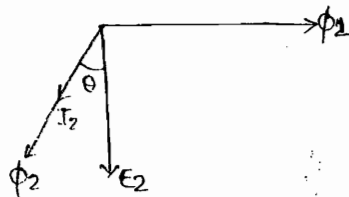
Pure inductive

$$\phi = 90^\circ$$

$$\cos 90^\circ = 0$$

$$T \propto E_2 I_2 \cos \phi_2$$





(38/30)

$$T \propto \frac{SV^2}{f}$$

$$\frac{N_s}{2} = \frac{120 f / 2}{P}$$

$$\frac{T_2}{T_1} = \frac{S_2}{S_1} \left(\frac{V_2}{V_1} \right)^2 \left(\frac{f_1}{f_2} \right)$$

$$\sqrt{S_2} = 2 S_1$$

slip speed constant

(44/31)

$$\text{Stator core loss} = 1.5k + 1200W = 2700W$$

$$\text{Rotor cu loss} = 900W$$

$$\text{mech. loss} = 1050W$$

$$P = \sqrt{3} V_L I_L \cos \phi - 2700W$$

(46/31)

$$S = \frac{1500 - 1410}{1500}$$

2000 → total

$$(1-S) 2000 \rightarrow 1880$$

$$P = I^2 R$$

$$I = \sqrt{\frac{P}{R}}$$

$$P = 1880W$$

(48/31)

$$V_1 = 400V, f_1 = 50, N = ?$$

$$V_2 = 240V, f_2 = 30Hz$$

$$\frac{V_1}{f_1} = \frac{V_2}{f_2}; V_2 = 240V$$

$$S_1 = \frac{1500 - 1440}{1500}$$

$$S_1 = 0.04$$

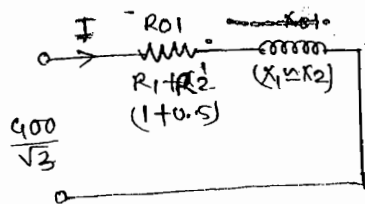
$$T \propto \frac{SV^2}{f}$$

$$\frac{T_2}{T_1} = \frac{S_2}{S_1} \left(\frac{V_2}{V_1} \right)^2 \left(\frac{f_1}{f_2} \right)$$

$$S_2 = 0.06$$

$$N = N_s(1-S)$$

$$= \frac{120 \times 30}{4} (1 - 0.06) = 840$$

(49)
32

$$T_g = \frac{60}{2\pi N_s} \cdot \frac{RCL}{s\pi}$$

$$= \frac{60}{2\pi N_s} \times 3I^2 R \times$$

$$T_g = \frac{3 \times 60}{2\pi N_s} \times \left\{ \frac{400/\sqrt{3}}{[1.5 + j2.4]} \right\}^2 \times 0.5$$

(50)
32

$$\eta = \frac{10000}{10000 + 1002 + 1354}$$

NL	BR
152A	72W
20A	?

(51)
32

$$\frac{T_{st}}{T_f} = 1.5$$

$$\frac{T_m}{T_f} = 3$$

$$\frac{T_{st}}{T_m} = \frac{1}{2} = \frac{(2sm)}{sm^2 + 1}$$

(61)
33

In all test watt-meter reading represent iron, friction & windage but iron losses are separated to represent as shunt branch R_c in the actual eq. ckt

(62)
33

$$T \propto \frac{S}{R_2}$$

$$R_2 \propto S$$

$$\frac{0.2S}{0.25 + 0.5} = \frac{0.03}{?}$$

(63)
33

$$\left. \begin{array}{l} 50\text{Hz} \rightarrow 1500 \\ 30\text{Hz} \rightarrow 900 \end{array} \right\} \begin{array}{l} \text{Possible speeds} \\ \text{may be} \\ 1500 \pm 900 \end{array}$$

(74)
34(75)
35

$$f = 48\text{Hz}$$

$$N_s = \frac{120(48)}{6}$$

$$N = 936$$